

PacketArch

The OT Traffic Simulation Platform for Industrial Security Teams

Executive Briefing

Version 1.5

May 2026

Why We Are Here

OT networks are the next attack surface — and the hardest one to defend confidently.

This briefing covers what PacketArch does, what it replaces, and what it returns to the business.

The OT Security Problem

Defenders are flying blind on the plant floor

- **No safe place to test.** Production lines cannot be the lab.
- **Real PLCs cost thousands** and ship with months-long lead times.
- **Generic packet tools don't speak OT** — no Modbus state, no vendor identity, no Cyber Vision fingerprint.
- **Detection tooling is unproven** until it has seen realistic ICS traffic **and** realistic attacks.
- **Tabletop exercises are theater.** Teams need traffic on the wire.

Result: millions invested in OT visibility tools that have never been validated end-to-end.

What PacketArch Is

A software platform that produces protocol-accurate industrial network traffic — on demand.

- Design plant networks visually, or describe them in plain English.
- Generate **packet-perfect** ICS traffic to PCAP, or inject it live onto a lab segment.
- Run **adversary playbooks** against your sensors and SOC — with after-action reports.
- Integrates natively with **Cisco Cyber Vision** for closed-loop validation.

One platform. Six verticals. Zero hardware required to get started.

Platform At A Glance

295

DEVICE TEMPLATES

18

OT VENDORS

7

ICS PROTOCOLS

6

INDUSTRY VERTICALS

6

ATTACK PLAYBOOKS

2

DEPLOYMENT MODES

Built for: SOC validation - red/blue team exercises - sensor onboarding - training - tool selection - vendor bake-offs.

Capability Matrix

Capability	What It Delivers
Visual Scenario Studio	Drag-and-drop plant networks, IEC 62443 zones & conduits
AI-Assisted Authoring	Natural-language scenario generation and review
Protocol Accuracy	Stateful engines for 7 ICS protocols, vendor-real fingerprints
Live Attack Simulation	6 kill-chain playbooks with per-action telemetry
After-Action Reporting	MITRE ATT&CK coverage, IOCs, packet-level evidence
Adaptive Traffic	Phase scheduling, time-of-day shaping, drift & jitter
Process Simulation	Realistic sensor values driven by physical models
Cyber Vision Integration	Bidirectional device matching and enrichment

Common Workflows

How teams actually use PacketArch

Workflow	Who runs it	Outcome
Sensor / Tool Validation	OT security ops	Prove Cyber Vision (or any DPI) sees what's actually on the wire — before trusting it in production
SOC Exercise (Red vs Blue)	SOC + IR teams	Run a kill-chain playbook against the live SOC, score detection coverage from the after-action report
Cyber Vision PoC Acceleration	Solutions engineering	Replace the 3-6 month "wait for the lab" with a Day-1 realistic site — close PoCs faster
Detection Engineering	Detection / threat-intel	Generate adversary ICS traffic on demand, tune Snort / Suricata / Splunk rules against ground truth
Training & Certification	Training / curriculum	Repeatable, resettable scenarios — every cohort runs against identical traffic, every time
Vendor Bake-Off	Procurement / TAC	Drive every candidate tool against one canonical scenario to compare apples-to-apples

Workflow Spotlight — End-to-End in a Day

Example: validate a new Cyber Vision sensor deployment

1. **Design** — pick the manufacturing template, AI-tweak it to match the customer's plant (5 minutes).
2. **Deploy** — start a live traffic agent on the lab segment Cyber Vision is monitoring (1 click).
3. **Match** — open the Cyber Vision Compare panel; confirm 100% MAC / 95% IP match across all devices.
4. **Enrich** — push vendor / model / firmware back into CV with one click.
5. **Stress** — run a `NETWORK_RECON` playbook to validate CV's anomaly detection lights up.
6. **Report** — download the after-action JSON for the customer's compliance file.

From empty install to evidence-grade report — single afternoon, single workstation.

Industry Verticals We Cover

Vertical	Representative Scenarios
Manufacturing	Assembly cells, robotic lines, SCADA over PROFINET / EtherNet-IP
Water & Wastewater	Treatment trains, lift stations, telemetry over Modbus / SNMP
Energy & Power	Substations, generation control, IEC-style telemetry
Oil & Gas	Wellhead control, custody transfer, pipeline RTUs
Building Automation	HVAC, access control, BACnet/IP estates
Transportation / ITS	Traffic cabinets, NTCIP signaling, roadside sensors

Every vertical ships with **pre-built templates**, **process models**, and **vendor-correct device libraries**.

Protocol Coverage

Protocol	Port(s)	Status
Modbus TCP	502	Production
EtherNet/IP (CIP)	44818 / 2222	Production
PROFINET	Layer 2	Production
S7comm / S7comm+	102	Production
BACnet/IP	47808	Production
SNMP / NTCIP	161 / 162	Production
OPC UA, DNP3, IEC 60870-5-104	Various	On roadmap

Each engine generates **startup, polling, and shutdown** sequences — not just packet shapes, but conversations.

AI-Augmented Authoring

Describe the plant. PacketArch builds it.

"Create a water treatment site with three pump stations, a Siemens S7-1500 master, and a Cisco IE switch in each zone."

- **Task-routed model selection** — Anthropic, OpenAI, or **Cisco CIRCUIT** per workload.
- **Scenario review** scores realism, conduit compliance, vendor accuracy.
- **AI Help** explains every screen and every recommendation in context.
- **Portable Scenario v1** — any LLM can author scenarios in a documented, importable JSON format.

AI is a feature flag. Air-gapped sites run identically with AI off.

Live Attack Simulation

Six end-to-end ICS adversary playbooks, executed against your real lab segment.

Playbook	Inspiration
TRITON-like	Safety system compromise
PIPEDREAM-like	Modular ICS toolkit
INDUSTROYER-like	Substation manipulation
HAVEX-like	OPC reconnaissance & exfil
Insider Threat	Trusted-user misuse
Network Recon	Pre-attack discovery

Plays out across the kill chain with real packets — recon, lateral movement, ICS-protocol abuse, C2 beaconing, impact.

After-Action Reports

Prove the exercise happened. Prove what the SOC saw — and what it missed.

- **Per-action telemetry** — packets emitted, targets hit, function codes used, registers touched.
- **MITRE ATT&CK coverage grid** — planned vs. fired techniques.
- **IOC capture** — attacker IPs, ports, SNMP communities, beacon patterns.
- **History on the scenario** — every run preserved for compliance and trend reporting.
- **Downloadable JSON** for SIEM correlation and tabletop evidence.

One click after each run produces an audit-grade record.

Cisco Cyber Vision Integration

Close the loop between simulation and your live OT visibility tool.

Feature	Outcome
Inventory pull	Compare PacketArch scenario to your real CV inventory
Device matching	MAC at 100% confidence, IP at 95%
Enrichment push	Send vendor, model, firmware back into CV
Fingerprint uniqueness	Per-instance serials prevent CV merging
Site identity rail	Consistent naming across CV centers and exercises

Validate that CV sees what you simulated — before you trust it in production.

Deployment Models

Mode	Use Case	Footprint
PCAP-only	Air-gapped labs, offline training, dataset generation	Single server, no agents
Live Agent	SOC validation, sensor tuning, red-team exercises	Server + lightweight Docker agents per segment

- **Same engines power both.** PCAP and live output are byte-for-byte equivalent for the same scenario.
- **One-script offline installer** with `docker save` images — works on disconnected sites.
- **Agents phone home** over WebSocket — no inbound firewall rules required.

Security & Compliance Posture

Area	Approach
Transport	TLS everywhere; self-signed by default, custom cert hot-swap supported
Secrets	Bcrypt-hashed agent tokens, Fernet-encrypted API keys
Setup	First-run wizard; no default admin password
Licensing	GPL-3.0, full third-party license manifest shipped
Air-Gap	Offline tarball, AI flag off by default, no outbound calls required
Data Residency	Self-hosted; nothing leaves the customer environment unless they enable cloud AI
IEC 62443	Conduit modeling and compliance checks built into scenario authoring

Economics — What PacketArch Replaces

Traditional Approach	Typical Cost	PacketArch Equivalent
Hardware testbed (12 PLCs + switches)	\$80K - \$250K	Software scenario, minutes to build
Custom traffic generator contract	\$150K+ / engagement	Built-in, unlimited reuse
Red-team ICS engagement	\$60K - \$200K	Repeatable playbooks, in-house
Cyber Vision PoC delays	3 - 6 months	Days, with realistic traffic on day 1
Training lab refresh	Annual hardware spend	Snapshot, clone, reset on demand

A single avoided testbed pays for **multi-year platform deployment** across all labs.

What An Engagement Looks Like

A typical 90-day journey from install to value

- 1. Install (Day 1)** — Offline tarball, one script, running on customer hardware.
- 2. First Scenario (Week 1)** — Import a vertical template or describe one in plain English.
- 3. Sensor Onboarding (Week 2-3)** — Generate traffic to Cyber Vision, validate inventory match.
- 4. First Exercise (Week 4-6)** — Run a kill-chain playbook against the SOC, review after-action report.
- 5. Operationalize (Week 8-12)** — Scheduled regression runs, training cohorts, vendor bake-offs.

No vendor lock-in. No data leaves the site. No production risk.

Roadmap

Theme	Items
Protocols	OPC UA, DNP3, IEC 60870-5-104
Attack Library	Expanded playbooks, custom action authoring
Reporting	Executive PDF exports, multi-run trend dashboards
Federation	Multi-lab scenario sync, central template registry
Tool Integrations	Splunk, Microsoft Sentinel, Dragos, Claroty connectors

Roadmap is **customer-driven** — verticals and protocol additions are prioritized by deployed sites.

Why PacketArch — In One Slide

- **Protocol-correct** where generic tools are not.
- **Vendor-realistic** down to MAC OUI and firmware string.
- **Repeatable** — same scenario, every time, every site.
- **Closed-loop** with Cyber Vision and your SOC tooling.
- **Operates air-gapped** — no cloud dependency required.
- **Auditable** — every attack run produces an evidence record.
- **Economical** — one platform replaces hardware labs, traffic contracts, and one-off red-team engagements.

Call to Action

Three ways to engage

1. **Executive demo** — 45 minutes, see a kill chain run end-to-end against a live SOC.
2. **Proof of value** — 30-day install in your lab, against your sensors, with your team.
3. **Strategic deployment** — multi-site rollout, training package, integration support.

Contact: github.com/ip-aegis/PacketArch

Thank You

PacketArch

Protocol-Accurate OT Traffic Simulation — for the teams defending what runs the physical world.

Version 1.5 May 2026